Word Embeddings

cat → [-, -, -]

2013

⇓

Transformer

Extra notes

Hey all is well → positive

well I am all happy → P

I am sad → N

(well, 2, 0)

(happy, 1, 0)

(I, 1, 1)

Tq. well I happy → positive

P 2 +1 +1 ⇒ 4 → Pos.

N 0 +1 +0 ⇒ [illegible]

I	am	happy	sad
1	1	1	0
1	1	0	1

happy = [1, 0]

→ TF-IDF →

Term Freq * Inverse Doc Freq

sentence level

vocab level

TF is for each word in a sentence

$$TF(w, d) = \frac{\text{\# time word occured in a sentence}}{\text{total \# of words in a sen}}$$

It was the best of times	age	and	it - - - -

$\frac{1}{6}$ $\frac{1}{6}$

IDF → $\log\left(\frac{\text{total no of docs}}{\text{no of docs in which this word occurs}}\right)$

foodishness → $\log\left(\frac{3}{1}\right)$

the → $\log\left(\frac{3}{3}\right)$ $\log(1)$

0

$tf_{the} * IDF_{the} = 0$

the	it
0	$tf_{it} * IDF_{it}$

cat → [1, 0, 1, 0, 0, 1]

cat → [0.4, 0.6, 0.8, 0.1, 0.2, 0.3]

love   like

cosine (   )

⇒ Recomendation system

↓

Matrix Factorization

$$A = B \cdot C^T$$

$C^T \rightarrow m \times d$

$B \rightarrow n \times d$

	w_1	w_2	w_3	w_4	w_5
w_1	0	2	0	–	→
w_2	–	–	–	–	–
w_3					
w_4					
w_5					

$$\begin{array}{c|c} & 1 \; \dots \; d \\ \hline u_1 & - - - - - \\ u_2 & \\ u_3 & \\ \vdots & \\ u_n & \end{array}$$

$\rightarrow B \times C^T$

$B \rightarrow w \times d$

w_1 w_2

$w_1 \to$ The

$w_2 \to$ cat

$w_3 \to$ dance

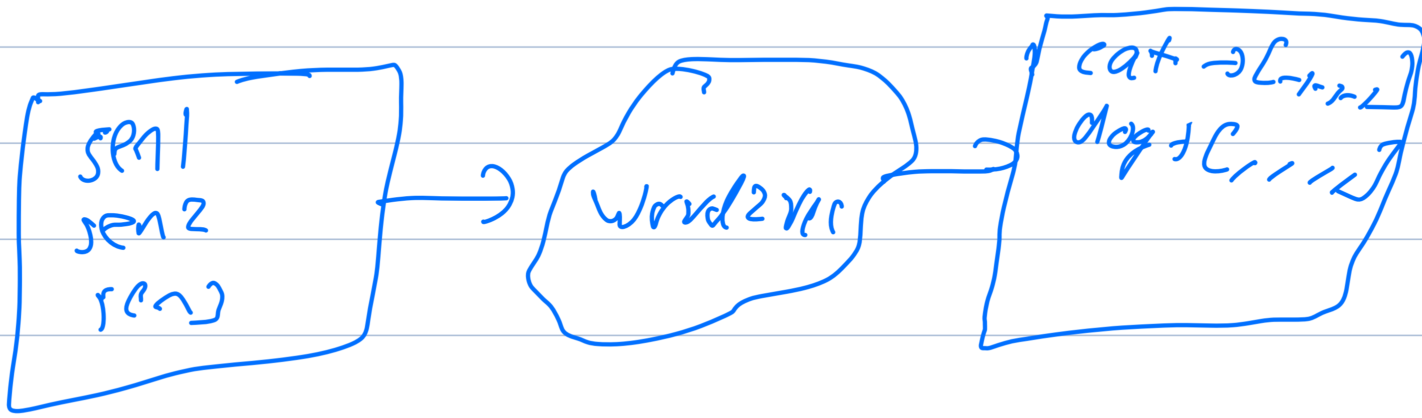
$w_4 \to$ sits

The cat dance

The cat sits

Word2Vec

cat → [0.1, 0.2, ...]



→ Shallow NN

→ word → 300

cat → [, , , , , ,]

word2vec → Semantic similarity

King → [0.2, 0.3, 0.4]
man → [0.1, 0.6, 0.1]
queen → [0.4, 0.2, 0.15]
woman → [0.2, 0.1, 0.7]

King + man = queen + woman

CBOW (Continous Bag of words

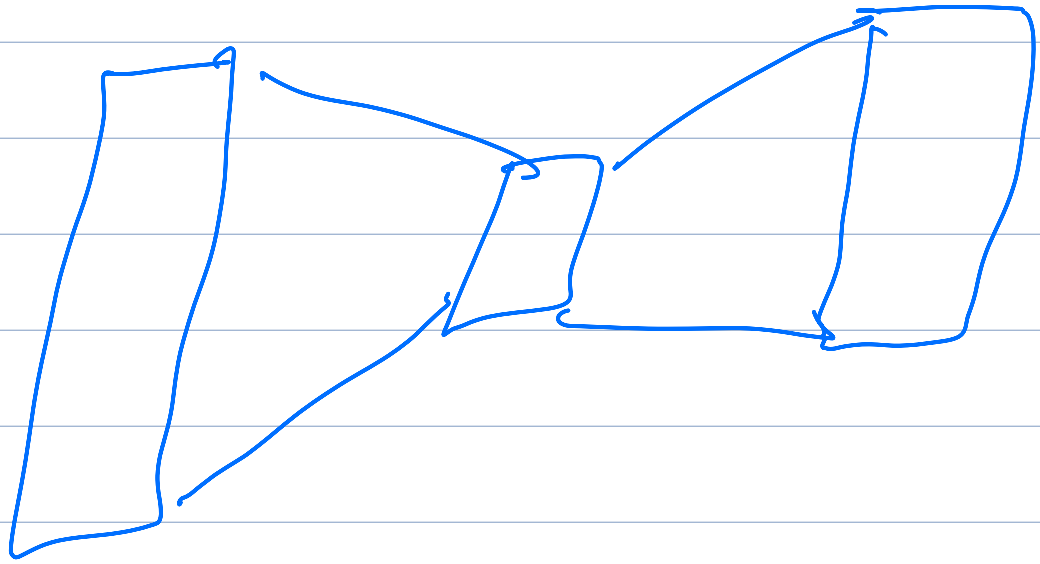
Paris is $\boxed{\text{capital}}$ of France

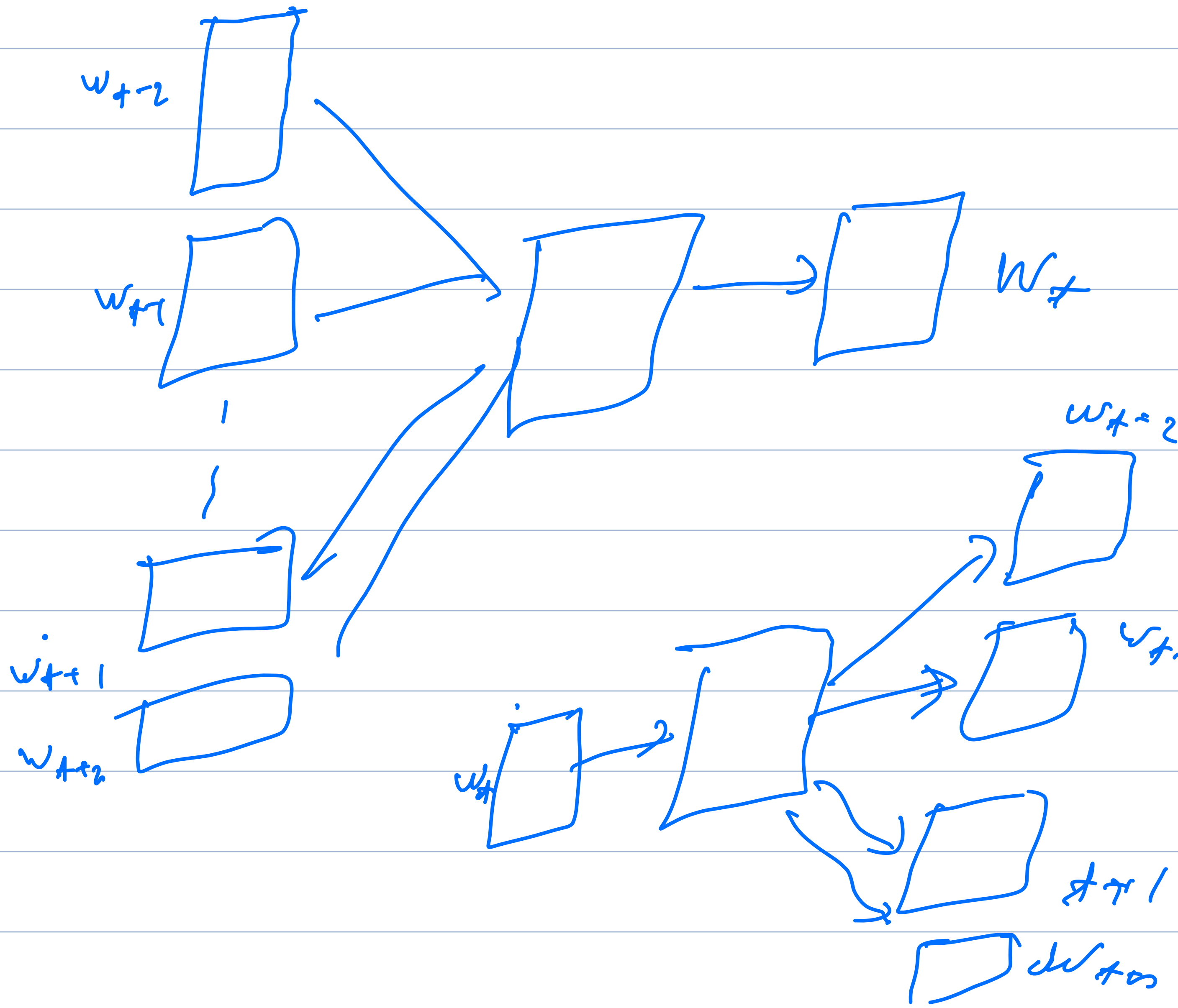
[Paris, is, of, France] → capital

Paris →
is →
of →
France →

Shallow NN → capital

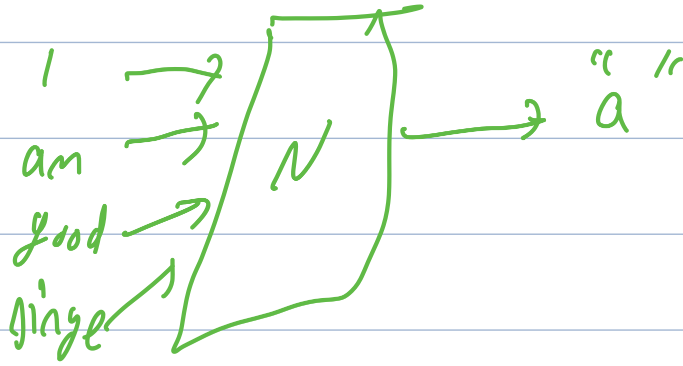
Auto encoder



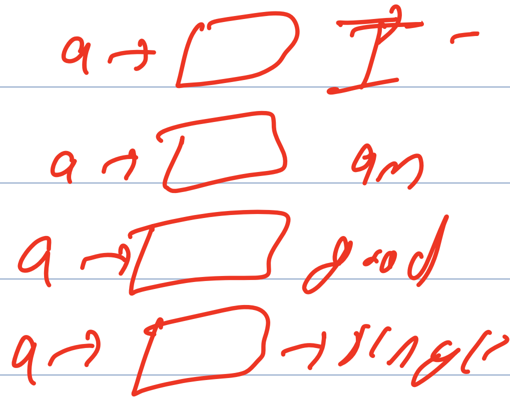


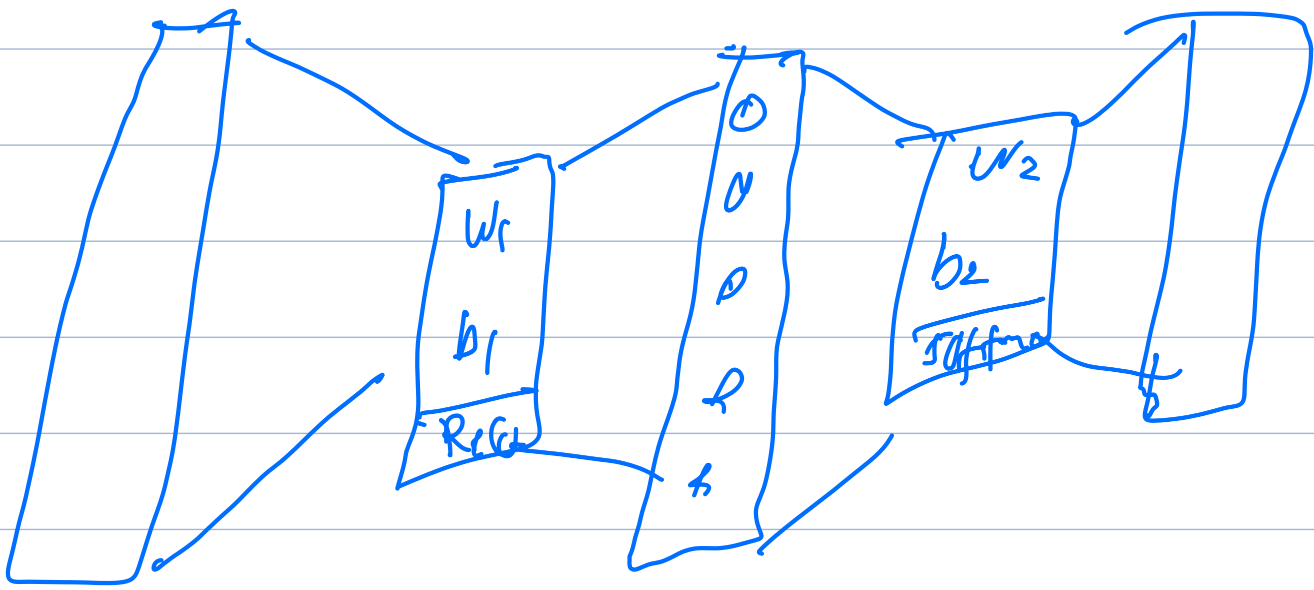
I am a good singer

CBOW



Skip-gram





hid

The cute cat jumps over the lazy dog

cw = 1

cute

jump

model → cat

→ corpus → I am happy bec I am learning

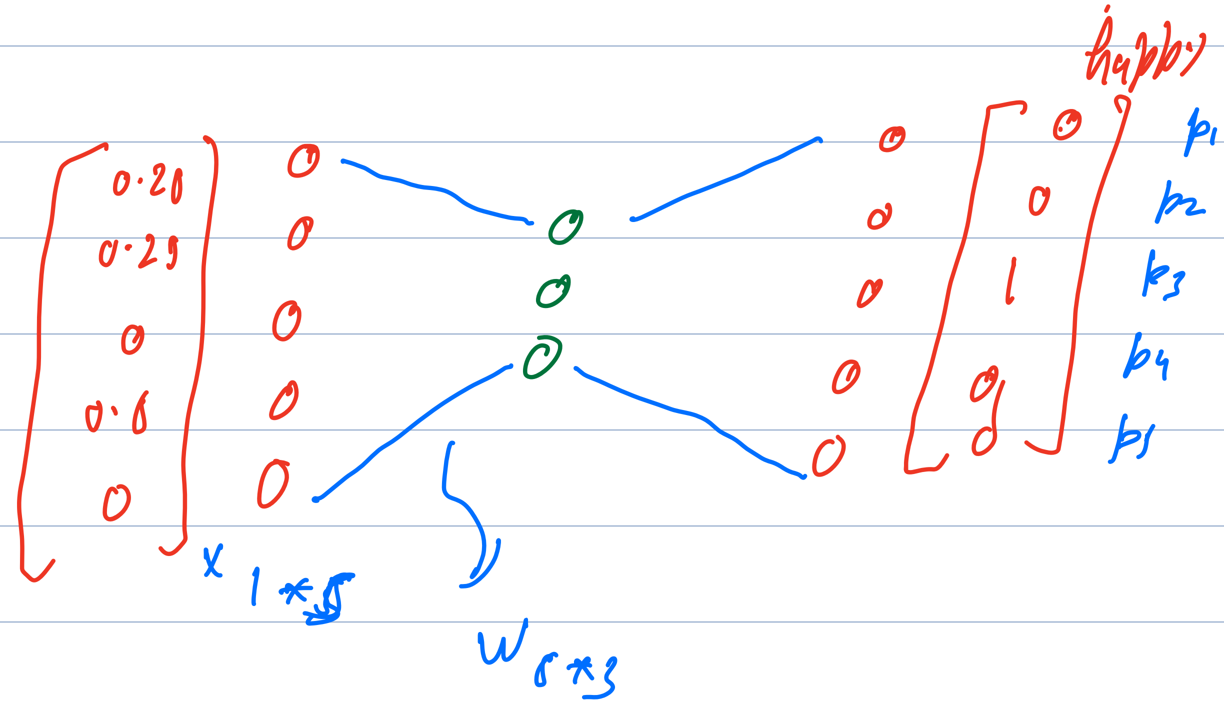
Vocab → am, because, happy, I, learning

$$
\begin{array}{c|c c}
 & \text{am} & \text{because} \\
\text{am} & 1 & 0 \\
\text{bec} & 0 & 1 \\
\text{happy} & 0 & 0 \\
\text{I} & 0 & 0 \\
\text{learning} & 0 & 0
\end{array}
$$

size of vocab

I, am, bec, I → happy

$$\left(\begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} + \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} \right) / 4 = \begin{bmatrix} 0.25 \\ 0.25 \\ 0 \\ 0.5 \\ 0 \end{bmatrix}$$



$$\mathrm{Relu}\,(X \cdot W + B)$$

Cross-entropy loss

$$J = -\sum_{K=1}^{V} y_K \log \hat{y}_K$$

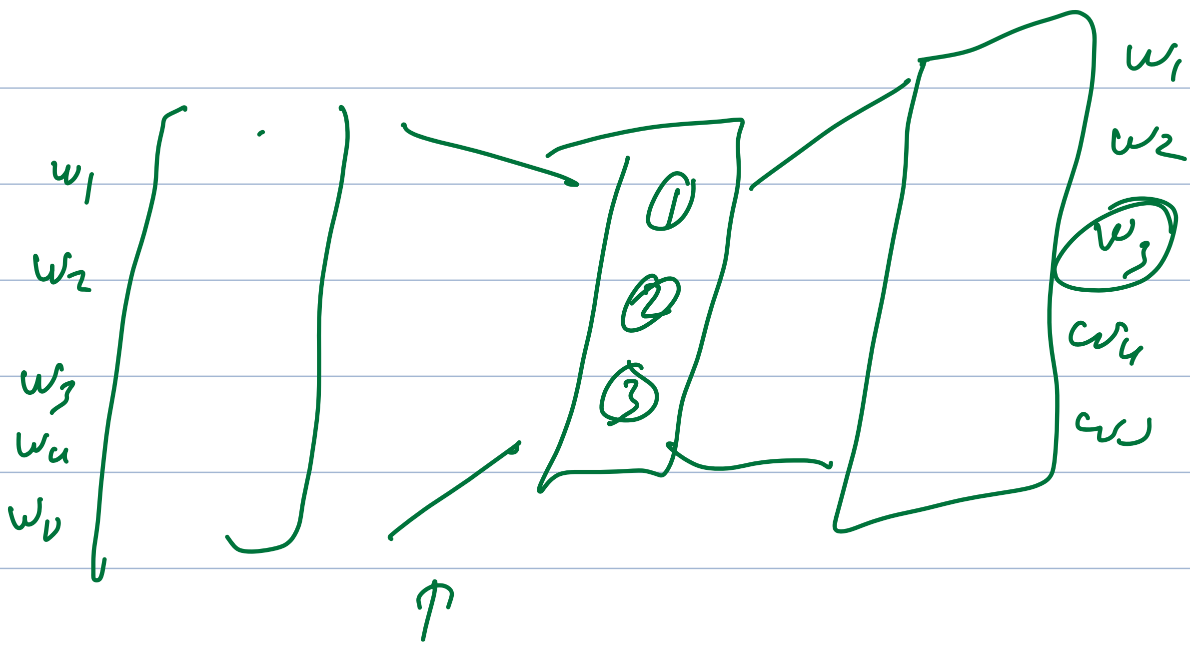
Actual $\Rightarrow \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_V \end{bmatrix}$ Pred $\Rightarrow \begin{bmatrix} \hat{y}_1 \\ \vdots \\ \hat{y}_V \end{bmatrix}$

$$y = \begin{array}{cl} \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} & \begin{array}{l} \text{am} \\ \text{bec} \\ \text{happy} \\ \text{I} \\ \text{learning} \end{array} \end{array} \qquad \hat{y} = \begin{bmatrix} 0.083 \\ 0.03 \\ 0.611 \\ 0.226 \\ 0.05 \end{bmatrix} \Rightarrow \log(\hat{y}) = \begin{bmatrix} -2.49 \\ -3.49 \\ -0.49 \\ -1.49 \\ -2.49 \end{bmatrix}$$

$$\Downarrow$$

$$y * \log(\hat{y})$$

$$J = 0.49 \leftarrow \begin{bmatrix} 0 \\ 0 \\ -0.49 \\ 0 \\ 0 \end{bmatrix}$$



$$W \rightarrow 5 * 3$$

$$\begin{array}{cc} & \begin{array}{ccc} 1 & 2 & 3 \end{array} \\ \begin{array}{c} w_1 \\ w_2 \\ w_3 \\ w_4 \\ w_5 \end{array} & \begin{bmatrix} & & \\ & & \\ & & \\ & & \\ & & \end{bmatrix} \end{array}$$